U^{2}ConvFormer: Marrying and Evolving Nested U-Net and Scale-Aware Transformer for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification plays an important role in the human exploration of the Earth. Recent research of deep learning-based HSI classification has been fast-growing, but still suffers from three obstacles: First, existing deep learning-based HSI works lack of extraction and utilization of multigrained multiscale information and multiscale local-to-global information. Second, most previous works have too fixed-sized receptive fields in their convolutional network parts to handle HSI classification problems, and pay no attention to the existence of asymmetries in the spectral-spatial dimension of the HSI data. Third, most networks for HSI classification are hand-craft. To this end, we propose a novel architecture in this article, which is the first to combine the advantages of nested U-Net and scale-aware Transformer, named U^{2}ConvFormer. Specifically, the nested U-Net structure can fully extract and aggregate multiscale spectral-spatial features at both inter- and inner stage granularity. The scale-aware Transformer takes multiscale local spectral-spatial features from the encoder of nested U-Net and produces multiscale global spectral-spatial features for its decoder. After that, we design a novel plug-and-play searchable operation called asymmetric spectral-spatial convolution (A2SConv), where asymmetric spectral-spatial feature pooling and multiscale feature extraction can be concurrently searched. Furthermore, we develop a customized search strategy to automatically design U^{2}ConvFormer, which uses advanced neural architecture search (NAS) methods to enable the customization of suitable models for different hyperspectral datasets. Experimental results on three benchmark datasets, including Indian Pines, Pavia University and Houston University 2018, validate the superiority of our proposed U^{2}ConvFormer, which achieves new state-of-the-art performance across different benchmark datasets.**

***Index Terms*— Nested U-Net, asymmetric spectral-spatial feature, hyperspectral image (HSI) classification, neural architecture search (NAS), scale-aware transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) classification is a valuable task in remote sensing, which is the basis for a large amount of practical applications including environmental sustainability [1], fine agriculture [2], and forest management [3], [4]. The objective is to classify each pixel into ground object classes utilizing its spectral and spatial information, which is preserved by hundreds of continuous narrow spectral channels.

In the early research phase, machine learning techniques, containing linear discriminant analysis (LDA) [10], random forest [11], and support vector machines (SVMs) [12], [13], [14] are usually utilized for HSI classification. The limitations of these traditional approaches lie in their strong reliance on prior knowledge, extracting only low-level features, and yielding less-than-optimal classification accuracy.

Recently, the superior performance in solving various computer vision problems with deep learning-based approaches has drawn the interest of HSI researchers, leading to a fast-growing development of convolutional neural network (CNN) and vision Transformer (ViT)-based HSI classification works For instance, kinds of CNN networks like 2-D CNN [15], 3-D CNN [5], [16], ResNet [17], and DenseNet [18] have been introduced into HSI classification. Unlike traditional machine learning algorithms, these CNN-based methods are able to extract high-level spectral-spatial features. To further handle ground objects of varying sizes, plenty of 2-D CNN and 3-D CNN methods employ multiscale strategy [18], [19] to solve HSI classification problems. Moreover, motivated by the outstanding performance of ViT in establishing long-distance associations in a wide range of image and video processing tasks [20], [21], [22], [23], plain ViT models are directly employed by some researchers to classify HSI data [6], [24]. Additionally, several HSI classification works that seek to leverage the strengths of both ViT and CNN are further proposed [8], [25]. Although above deep learning-based methods have achieved promising results, they still suffer from three crucial issues.

First, few works in HSI classification consider extracting and utilizing multigrained multiscale information and multiscale local-to-global information. For CNN-based methods, they are limited by their receptive field and lack the ability to model long-distance associations. To alleviate this, multiscale techniques are applied. However, commonly used multiscale techniques ignore two important factors that affect HSI classification. On the one hand, some networks have a significant reduction in resolution as depth increases [19], but HSI data often has mall and narrow ground objects and maintaining high resolution in HSI feature extraction is important. On the other hand, although some methods maintain high

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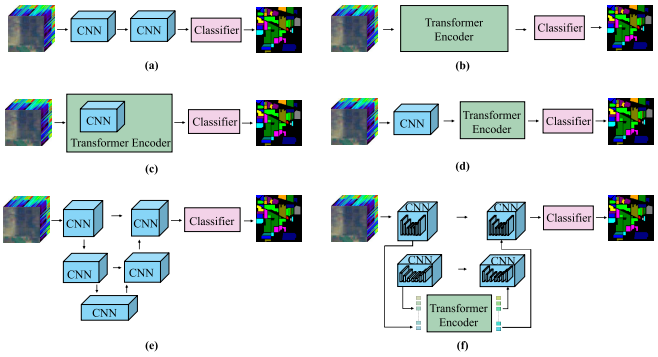


Fig. 1. Overview of existing well-recognized backbone networks for HSI classification task. (a) CNN-based network [5]. (b) ViT-based network [6]. (c) CNN and ViT combined network with multiscale CNN inside multihead attention module of ViT [7]. (d) CNN and ViT combined network with multiscale CNN in the front and ViT in the back [8]. (e) Multiscale CNN-based network via U-Net architecture [9]. (f) Our U^{2}ConvFormer architecture.

resolution features by extracting and fusing multiscale features of multiple network stages [18], inner stage multiscale features are usually ignored. For ViT-based methods, they perform excellently in modeling long-distance associations but lack competence in acquiring local information. Moreover, they only take the output tokens of the previous layer as input and produce tokens for the next layer, which is lack of utilization of hierarchical features. For CNN and ViT combined methods, although they attempt to take advantage of both ViT and CNN models [8], [25], they only make use of single-scale features without consideration of possible differences in shapes and sizes of ground objects. Very recently, some HSI researchers begin to take into account the combination of multiscale CNN architecture and ViT model [7], [26]. However, their designs are either multiscale CNN inside multihead attention module of ViT or multiscale CNN in the front and ViT in the back, ignoring intrastage extraction of multiscale local-to-global information. We show the sketch maps of aforementioned works in Fig. 1.

Second, for most existing CNN-based and CNN-and-ViT-combined HSI classification methods, their ability to classify HSI data is limited by the fixed-sized receptive field of the CNN parts. Since most previous works pay no attention to the existence of asymmetries in the spectral-spatial dimension of the HSI data, it would be helpful to use operations that enable asymmetric information extraction rather than spectral-spatial symmetric operations. Moreover, as mentioned above, the ground objects in HSI data have a wide variety of sizes and shapes, thus, it is worth providing variable and flexible receptive fields for each layer of each stage of CNN.

Third, most networks for HSI classification are hand-craft. As previously analyzed, tailoring the receptive fields for each layer of the network is worthwhile. However, manually selecting an appropriate operation for each layer is almost impossible when facing a combination of so many candidate operations. Meanwhile, customized processes are required as physical characteristics vary between HSI datasets acquired by different sensors [27]. With this in mind, we enlist the help of neural architecture search (NAS) to automate the network design process [28], cutting down the necessary effort and domain knowledge. Early NAS approaches utilize evolutionary algorithm [29], [30] and reinforcement learning [28], but usually incur massive computation. Compared with other NAS strategies, differentiable architecture search (DARTS) [31] first formulate the search space as a continuous domain where the probability of employing an operator is represented by the importance weight, and then search the optimal architecture using gradient descent, which significantly decrease the complexity of NAS tasks. Moreover, Beta-Decay regularization is proposed in [32] to improve DARTS-based methods without extra changes or cost, which is adopted in this article and verified in HSI classification tasks.

In this work, we propose a novel HSI classification network named U^{2}ConvFormer, which combines the advantages of nested U-Net and scare-aware Transformer for the first time. In detail, U^{2}ConvFormer consists of two parts: a well-designed U^2-Net that includes a two-layer encoder and a two-layer decoder and a scale-aware Transformer. The scale-aware Transformer takes the reshaped multiscale local spectral-spatial features as input and produces scale-aware global spectral-spatial features. Each layer of encoder or decoder is a U-Net architecture consisting of asymmetric spectral-spatial convolution (A2SConv) layers. A2SConv is a novel plug-and-play asymmetric spectral-spatial searchable operation. Specifically, the asymmetric pooling in A2SConv maximally extracts the primary features of HSI data. And the receptive field of the 3-D convolution in A2SConv is designed to be searchable rather than fixed-sized. Finally, we adopt the advanced NAS strategy β-DARTS to automatically design U^{2}ConvFormer, searching for the appropriate A2SConv in different layers of encoder and decoder as well as the best number of heads and MLP ratio in scale-aware Transformer. Note that the A2SConv is proposed and studied in our recent conference paper (i.e., ICASSP2023 oral) [27]. In this work, we further focus on exploring the searchable architecture and propose a novel searchable structure consisting of nested U-Net and scale-aware Transformer. More experiments are done and better performance is achieved.

Our contribution to this work can be summarized as follows.

1) We propose a novel architecture named U^{2}ConvFormer. The nested U-Net structure can fully extract and aggregate multiscale spectral-spatial features at both inter- and inner stage granularity. The scale-aware Transformer takes the reshaped multiscale local spectral-spatial features as input and produces scale-aware global spectral-spatial features. To the best of our knowledge, it is the first time to combine the advantages of nested U-Net and scare-aware Transformer to construct a model to classify HSI data.
2) We then design a novel lightweight and plug-and-play searchable operation called A2SConv. To address the asymmetries in the spectral-spatial dimension of the HSI data and objects with a wide variety of sizes and shapes, a joint outer search space and inner search space is employed, where asymmetric spectral-spatial feature pooling and multiscale feature extraction can be concurrently searched. A2SConv can be flexibly built into U^{2}ConvFormer.
3) We further develop an search strategy to automatically design U^{2}ConvFormer, which uses advanced NAS methods to enable the customization of suitable models for

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different hyperspectral datasets. Extensive experiments are conducted on three widely used HSI benchmarks including Indian Pines, Pavia University, and Houston University 2018. Models produced by our method achieve competitive classification results compared with other state-of-the-art methods.

We organize the remaining section of this article as follows. In Section II, we revisit the related works about hand-craft and NAS-based deep learning networks for HSI classification. Then, Section III gives a detailed illustration of the proposed U²ConvFormer. Section IV shows the comprehensive experiments including comparisons with other HSI classification approaches and ablation studies. Finally, Section V presents the conclusions.

II. Related Works

A. Hand-Craft Networks for HSI Classification

1) CNN-Based Networks: In past decade, kinds of CNN-based networks like 2-D CNN [15], 3-D CNN [5], [16], ResNet [17], and DenseNet [18] have been widely used in deep learning-based HSI classification task [16], [33], [34], [35]. Further, various multiscale strategies are employed in these CNN networks for handling ground objects of varying sizes in HSI datasets. MS-Densenet [18] utilizes dense blocks to fuse multiscale features among different layers. SC-EADNet [19] proposes a lightweight HSI classification network that utilizes multiscale CNN for contextual information extraction.

2) ViT-Based Networks: Transformer [36] is an advanced model that has been first used in natural language processing and then successfully adapted to lots of computer vision tasks [20], [22], [23]. Inspired by the success of Transformer-based vision models, Hong et al. [6] train the plain ViT model for solving HSI classification problems. Two more strategies, named group embedding and cross-layer connectivity, are proposed to adapt to the characteristics of HSI data. On one hand, ViT-based networks take the output tokens of the previous layer as input and produces tokens for the next layer, which is lack of utilization of hierarchical features. Compare with it, the scale-aware Transformer in U²ConvFormer takes the reshaped multiscale local spectral-spatial features as input, where these features are produced by different layers of the encoder of the nested U-Net and concatenated along the channel dimension. Further, the learned scale-aware global spectral-spatial features are injected with the corresponding local features to augment the representation of HSI cubes. On the other hand, in order to evolve the optimal architecture of the scale-aware Transformer, we leverage the power of advanced NAS technology to search for the number of heads in the multihead self-attention (MHSA) block and the MLP ratio in the feed-forward network (FFN) block.

3) CNN and ViT Combined Networks: Recently, many efforts have been devoted to leverage the strengths of both ViT and CNN. HiT [25] catches small spectral differences and extracts local spatial background features by embedding convolution operations into the Transformer structure. In [8], a Gaussian weighted feature tokenizer is designed to process local spectral-spatial features obtained from the CNN, which generates tokens that are then fed into the Transformer module for further feature extraction. In addition, some multiscale feature extraction techniques are integrated into CNN and ViT combined networks to obtain better contextual information. In [7], a multiscale feature extraction module is designed and inserted into the Transformer-based network, and a feature fusion module is proposed to aggregate cross-level features. SSFE-Former [26] proposes a mixture structure of multiscale CNN in the front and ViT in the back, where CNN takes multiscale HSI cubes and generate multiscale token embeddings for ViT to obtain better spectral-spatial features. The LESSFormer [37] designs the HSI2Token module and the local-enhanced transformer encoder to adaptively produce spectral-spatial tokens and enhance local information while preserving global information, respectively. GAHT [38] augments the extraction of HSI local features by proposing a grouped pixel embedding module within the Transformer framework. 3DSwinT-HCL [39] designs a multiscale local contrastive learning module to explore 3DSwinT-based hierarchical contrastive learning method, which enhances pixel-level representation of HSI.

In computer vision, a number of works also pay attention to the design of CNN-and-ViT-combined models [40], [41] and multiscale CNN feature extraction methods [42], [43], [44]. Among them, Topformer [41] feeds tokens of different scales into Transformer to obtain scale-aware global semantic features, which are fused with the related local features to enhance representation for semantic segmentation. U²-Net [44] design a two-level nested U-Net structure for salient object detection, and the top level of which is a U-Net structure while the bottom level of which consists of U-Net-like blocks, which can extract and aggregate multiscale features at multiple granularities. Different from them, we are the first to combine the advantages of nested U-Net and scale-aware Transformer to design a new architecture named U²ConvFormer, and are the first to evolve it to classify HSI data. Specifically, we design a novel asymmetric spectral-spatial searchable operator A2SConv to build the U²ConvFormer architecture, and use advanced NAS methods to enable the customization of suitable architectures for different datasets.

B. Neural Architecture Search for HSI Classification

The purpose of NAS is to automate the design process of neural networks [28], [29], [30], [45], [46]. Despite the favorable search performance achieved by the previous reinforcement learning [28] and evolutionary algorithm [29], [30]-based NAS methods, the computational and time costs of evaluating each network individually are so high that a large amount of computational resources is required to be consumed. Recently, supernet-based NAS has greatly reduced the search cost by training a supernet that contains all candidate networks. Among them, DARTS [31] is widely used because it implements supernet training and architecture search simultaneously through bilevel optimization, which further reduces the search cost. However, DARTS also encounters the tough problems of search robustness and architectural generalization in searching progress. To address the problem, P-DARTS [47] and DARTS+ [48] directly restrict the number of skip connections to relieve the performance degradation, while DOTS [49]

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decouples the topology representation from operation weights and makes an explicit topology search. Compared with these DARTS-based automatic strategies, β-DARTS [32] addresses the two tough problems of DARTS at the same time without changing the searching progress or adding extra cost or datasets. Therefore, it has advantages in simplicity, efficiency and effectiveness.

In recent years, some NAS-based HSI classification methods have been proposed to automatically design neural networks with excellent performance, such as particle swarm optimization-based CPSO-Net [50] and DARTS-based 1-D Auto-CNN and 3-D Auto-CNN [51]. Despite the improvement in classification accuracy, these methods lack the design of search space with respect to the data characteristics of HSI. The recently proposed NAS-based HSI classification methods have investigated more on customized search space [52], [53], [54], [55]. EB-CNN [52] uses the proposed generic convolution block and generic pooling block to form the search space and search with genetic algorithms. SSTN [53] applies the attention mechanism and designs the spatial attention module and spectral association module, as well as proposes a factorized architecture search framework. 3D-ANAS [54] explores the design of asymmetric decomposition convolution, and HyT-NAS [55] further applies it to spatial-dominated cells and spectral-dominated cells as well as searches for a CNN-ViT-combined network.

Compared to these NAS-based methods, our U^{2}ConvFormer has three main differences.

1) The search space built on nested U-Net and scale-aware Transformer. In this work, we search for the appropriate A2SConv in different layers of encoder and decoder of nested U-Net, as well as the best number of heads and MLP ratio in the scale-aware Transformer.
2) The asymmetric spectral-spatial searchable operator A2SConv. The asymmetric pooling maximally extracts the primary features of HSI data. And the receptive field of the 3-D convolution in A2SConv is designed to be searchable rather than fixed-sized.
3) The search strategy. We adopt the advanced NAS strategy β-DARTS that can ensure the search robustness and architectural generalization to automatically design U^{2}ConvFormer for different datasets.

III. Proposed Method

As shown in Fig. 2, the proposed U^{2}ConvFormer aims to search for the suitable A2SConv in different layers of encoder and decoder of nested U-Net, as well as the best number of heads and MLP ratio in scale-aware Transformer, with the advanced NAS strategy β-DARTS. In this section, we will first illustrate the two key architectures of U^{2}ConvFormer: the nested U-Net and the scale-aware Transformer. Then, we will introduce the well-designed search space of U^{2}ConvFormer, where asymmetric spectral-spatial feature pooling and multiscale feature extraction can be concurrently searched. Finally, we introduce the advanced NAS strategy.

A. U^{2}ConvFormer

First, we feed cubes of size $n \times n \times$ Band that are randomly cropped on the HSI dataset into the U^{2}ConvFormer (n is chosen via the experimental results in Section IV). Specifically, the U^{2}ConvFormer consists of two parts: a well-designed nested U-Net that includes an m-layer encoder and an m-layer decoder (m is set to 2 via the experimental results in Section IV), and a scale-aware Transformer.

1) Nested U-Net: The nested U-Net architecture in our U^{2}ConvFormer is a well-designed U^2-Net consisting of a two-layer encoder and a two-layer decoder. Each layer of the encoder or decoder is a U-Net architecture consisting of A2SConv layers for capturing intrastage multiscale features. Formally, the input feature of each layer $X_{\text{in}} \in (C_{\text{in}}, S, H, W)$ is first transformed by a plain 3-D convolution layer, denoted as $B(X_{\text{in}})$. Then, a U-Net architecture consisting of d A2SConv layers (d is usually configured according to the resolution of the input feature maps) takes the intermediate feature map $B(X_{\text{in}}) \in (C_{\text{out}}, S, H, W)$ as input and produces intrastage multiscale information $U(B(X_{\text{in}})) \in (C_{\text{out}}, S, H, W)$. At last, the intrastage multiscale information is fused with the local feature map $B(X_{\text{in}})$. In conclusion, the output of each layer of the encoder or decoder is defined as the first equation of

$$\begin{aligned} X_{\text{out}} &= \underbrace{U(B(X_{\text{in}}))}_{\text{multiscale feature}} + \underbrace{B(X_{\text{in}})}_{\text{local feature}} \\ X_{\text{out}} &= \underbrace{F_2(F_1(X_{\text{in}}))}_{\text{local feature}} + \underbrace{X_{\text{in}}}_{\text{original feature}} \end{aligned} \tag{1}$$

where $B()$ represents a plain 3-D convolution layer, $U()$ defines the U-Net architecture consisting of d A2SConv layers, and $F_1()$ and $F_2()$ stands for the convolution layers of common residual block.

When comparing the residual U-block with the original residual block [56] [denoted as the second equation of (1)], the primary difference can be summarized as follows. First, a U-Net structure consisting of A2SConv layers is employed instead of the plain convolution. Second, the original feature of residual connection is replaced by the local feature produced by the basic block (i.e., the plain 3-D convolution layer). As such, the inner stage multiscale features can be acquired with less essential feature loss by each layer of the encoder or decoder. Further, thanks to the progressive downsampling, progressive upsampling, and feature concatenation in the U-Net architecture, the nested U-Net can fully extract and aggregate multiscale spectral-spatial features at both inter- and inner stage granularity.

2) Scale-Aware Transformer: The scale-aware Transformer consists of L blocks, each of which includes the MHSA block, the FFN and residual connections. The multiscale local spectral-spatial features from the two-layer residual encoder of the nested U-Net are concatenated and sent to scale-aware Transformer to obtain multiscale global spectral-spatial features. More specifically, the multiscale local spectral-spatial features $\{X_l^i \mid X_l^i \in (C_{\text{out}}, S, H, W)\}_{i=1}^m$ are first reshaped and transposed to $\{X_l^i \mid X_l^i \in (N, H, W)\}_{i=1}^m$ (m indicates the number of scales) and then computed by three 1×1 convolution layers to map three feature vectors named query (Q), key (K) and value (V). In order to speed up runtime, we follow the settings of TopFormer [41] to abandon the Layer normalization layer and attach a batch normalization to each convolution

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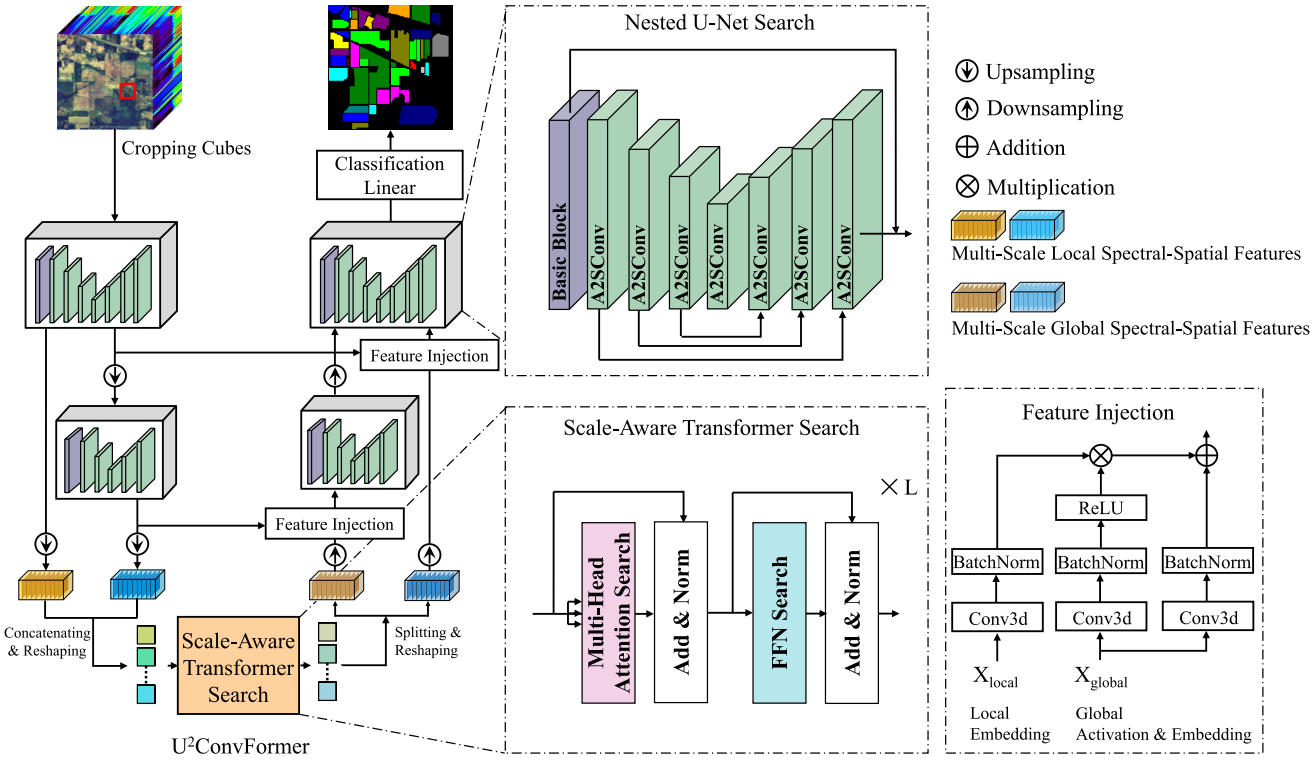


Fig. 2. Overview of the proposed U²ConvFormer. First, cubes of size $n \times n \times$ band that are randomly cropped on the HSI dataset are fed into the searchable structure of two-level nested U-Net with scale-aware Transformer, which includes a two-layer residual encoder, a two-layer residual decoder, and a scale-aware Transformer. Second, the encoder takes HSI cubes as input and produces multiscale local spectral-spatial features. The scale-aware Transformer takes the reshaped multiscale local spectral-spatial features as input and produces scale-aware global spectral-spatial features. The decoder takes the injected multiscale global and local features as input and produces augmented features. Finally, the classification linear layer takes the output of the final stage and produces the final classification results. The detailed illustration of the U² encoder and decoder search is shown above the middle column. Each layer of the encoder and decoder is of residual structure and dominated by the A2SConv, which is a novel plug-and-play asymmetric spectral-spatial searchable operation. A detailed configuration of the A2SConv is given in Fig. 3. The architecture of the proposed searchable scale-aware Transformer is shown below the middle column, which consists of L same blocks. A detailed configuration of it is given in Fig. 4.

operation. Further, the MHSA mechanism is computed as follows:

$$\begin{aligned} \text{Attention} &= \text{softmax}(QK^T)V \\ \text{MHA} &= \sigma(\text{BN}(\text{Attention})). \end{aligned} \tag{2}$$

Inspired by [57], the FFN block in our scale-aware Transformer consist of an 1×1 convolution layer, a deep-wise convolution layer, and an 1×1 convolution layers. In conclusion, the output of the scale-aware Transformer is defined as follows:

$$X_g = \text{MLP}(\text{DWConv}(\text{MLP}(\text{MHSA} + X_l))) + \text{MHA} + X_l \tag{3}$$

where X_g stands for a series of multiscale global spectral-spatial features, represented as $\{X_g^i \mid X_g^i \in (N, H, W)\}_{i=1}^N$. DWConv means the deep-wise convolution layer, MLP is the 1×1 convolution layer, and MHSA represents the MHSA mechanism. The number of Transformer blocks L, the number of heads, and the MLP ratio will be illustrated in Section III-B.

As shown in Fig. 2, the scale-aware Transformer takes the multiscale local spectral-spatial features from the two-layer residual encoder of the nested U-Net as input. Then, it exchange information among the local spectral-spatial features of different scales and consequentially obtain multiscale global spectral-spatial features. After that, the series of multiscale global spectral-spatial features are concatenated with the series of multiscale local spectral-spatial features to obtain multiscale local-to-global spectral-spatial features, which are then fed to the two-layer residual decoder of the nested U-Net. Finally, the classification linear layer take the output of the final stage and produces the classification results.

B. Search Space

For evolving the architecture U²ConvFormer for HSI classification tasks, we design a novel search space that achieves the search of A2SConv in different layers of encoder and decoder of the nested U-Net as well as the search of head number and MLP ratio in the scale-aware Transformer.

1) A2SConv Search: In U²ConvFormer, each encoder or decoder is designed as a U-Net architecture consisting of A2SConv layers. A2SConv is a novel plug-and-play asymmetric spectral-spatial searchable operation. Specifically, the outer search space (shown in Fig. 3 masked with pink color) searches for an asymmetric spectral-spatial feature pooling. Concurrently, the inner search space (shown in Fig. 3 masked with gray color) searches for 3-D convolutions with multiscale receptive fields. Thanks to this design, the feature extraction produced by A2SConv breaks the restrictions of fixed-sized convolutional kernels, and maximally extracts the primary features of HSI data.

2) Outer Search Settings: The outer search space is designed to minimize the computational cost in handling high-dimensional HSI data as well as maximally extracts

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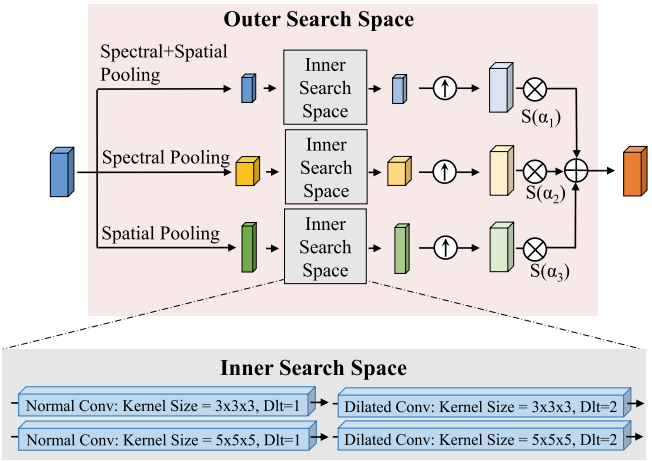


Fig. 3. Architecture of our searchable A2SConv. The outer search space (masked with pink color) searches for an asymmetric spectral-spatial feature pooling which maximally extracts the primary features of HSI data. The inner search space (masked with gray color) concurrently searches for 3-D convolutions with multiscale receptive fields.

the primary features of HSI data. It contains asymmetric spectral-spatial feature pooling and upsampling. In terms of pooling operations, three different candidates are considered: spectral pooling utilizing average pooling with stride $(s, 1, 1)$, spatial pooling utilizing average pooling with stride $(1, s, s)$, and spectral-spatial pooling utilizing average pooling with stride (s, s, s), respectively (s is set to 2 according to the size of HSI cubes). Furthermore, instead of designing the same fixed operations in all pooling layers of the network architecture (i.e., pooling operations in the spectral dimension or spatial dimension or the whole dimension), we search for the customized pooling operation for each A2SConv of the each layer of encoder or decoder in U^{2}ConvFormer architecture.

3) Inner Search Settings: To address the issue of varying sizes and shapes of ground objects in HSI data, the proposed inner search space searches for 3-D convolutions with multiscale receptive fields. The four different candidates are designed considering receptive field dominated by dilatation rate (d) and kernel size (k): Conv3d ($k = 3$, $d = 1$), Conv3d ($k = 3$, $d = 2$), Conv3d ($k = 5$, $d = 1$), and Conv3d ($k = 5$, $d = 2$).

4) Scale-Aware Transformer Search: Different from the plain ViT model [20], our scale-aware Transformer takes the reshaped multiscale local spectral-spatial features as input and produces multiscale global spectral-spatial features. To further evolve the architecture of the scale-aware Transformer, we design a search space on it, which searches for the number of heads in the MHSA block and the MLP ratio in the FFN block.

5) MHSA Search: To better exchange information among the spectral-spatial features of different scales and obtain better multiscale global spectral-spatial features, the number of heads is designed as a searchable parameter. As shown in Fig. 4, the multiscale local spectral-spatial features are fed to three candidate paths corresponding to different number of heads, consisting of $\{1, 2, 4\}$.

6) FFN Search: To reduce the computational cost, the MLP ratio of the FFN block is designed as a searchable parameter. As shown in Fig. 4, the output features of the MHSA block

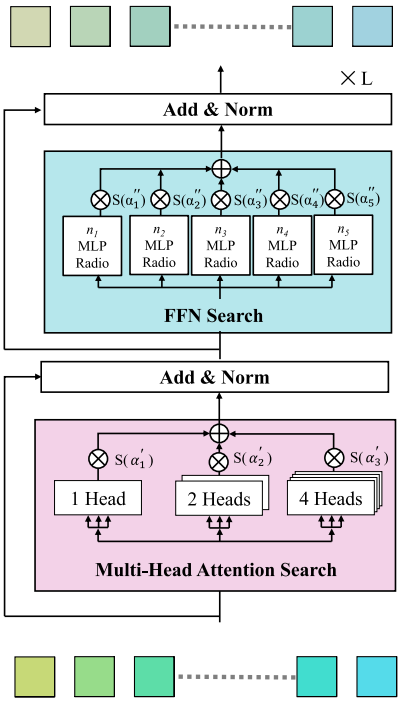


Fig. 4. Architecture of our searchable scale-aware Transformer, which searches for the number of heads in the MHSA block and the MLP ratio in the FFN block.

are fed to five candidate paths corresponding to different MLP ratios, consisting of $\{n_1, n_2, n_3, n_4, n_5\}$. $\{n_1 - n_5\}$ is set to $\{0.6, 0.7, 0.8, 0.9, 1.0\}$ and most of them is smaller than the plain MLP ratio in [41].

C. Search Strategy

We adopt the advanced NAS method β-DARTS [32] to automatically design U^{2}ConvFormer, searching for the appropriate A2SConv in different layers of the encoder and decoder of nested U-Net, as well as the best number of heads and MLP ratio in scale-aware Transformer. Each A2SConv block in each layer of nested U-Net has a learnable parameter α_k. Each MHSA block or FFN block in scale-aware Transformer has a corresponding learnable parameters α'_k or α''_k. The output of MHSA block or the FFN block or the A2SConv block is formulated as follows:

$$\overline{O}(x)^{(i)} = \sum_{k=1}^{|O|} F\left(\alpha_k^{(i)}\right) O_k(x)$$
$$F\left(\alpha_k^{(i)}\right) = \frac{\exp\left(\alpha_k^{(i)}\right)}{\sum_{k'=1}^{|O|} \exp\left(\alpha_{k'}^{(i)}\right)} \tag{4}$$

where x stands for the input of each block, $F(a_k^{(i)})$ is the softmax activation of the corresponding parameter $\alpha_k^{(i)} \in \{\alpha_k^{(i)}, \alpha_k'^{(i)}, \alpha_k''^{(i)}\}$, and $\overline{O}(x)^{(i)}$ stands for the mixed output. For the A2SConv block, O is the candidate operator set. For the multihead self-attention block, O is the set of self-attention with different numbers of heads. For the FFN block, O is the set of FFN with different MLP ratios.

Noted that in our proposed search strategy, the learnable parameters are not shared across layers within either the

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TABLE I
LAND COVER TYPE, THE QUANTITY OF LABELED TRAINING SAMPLES, AND THE NUMBER OF TESTING SAMPLES FOR THE INDIAN PINES, PAVIA UNIVERSITY, AND HOUSTON UNIVERSITY 2018 DATASETS

	Indian Pines			Pavia University			Houston University 2018		
Class	Land Cover Type	Train	Test	Land Cover Type	Train	Test	Land Cover Type	Train	Test
1	Alfalfa	5	41	Asphalt	332	6299	Healthy grass	490	9309
2	Corn-notill	143	1285	Meadows	932	17717	Stressed grass	1625	30877
3	Corn-mintill	83	747	Gravel	105	1994	Artificial turf	34	650
4	Corn	24	213	Trees	153	2911	Evergreen trees	680	12915
5	Grass-pasture	48	435	Metal sheets	67	1278	Deciduous trees	251	4770
6	Grass-trees	73	657	Bare Soil	251	4778	Bare earth	226	4290
7	Grass-pasture-mowed	3	25	Bitumen	67	1263	Water	13	253
8	Hay-windrowed	48	430	Bricks	184	3498	Residential buildings	1989	37783
9	Oats	2	18	Shadows	47	900	Non-residential buildings	11188	212564
10	Soybean-notill	97	875	-	-	-	Roads	2293	43573
11	Soybean-mintill	245	2210	-	-	-	Sidewalks	1701	32328
12	Soybean-clean	59	534	-	-	-	Crosswalks	76	1442
13	Wheat	20	185	-	-	-	Major thoroughfares	2317	44031
14	Woods	126	1139	-	-	-	Highways	493	9372
15	Buildings-Grass -Trees-Drives	39	347	-	-	-	Railways	347	6590
16	Stone-Steel-Towers	9	84	-	-	-	Paved parking Lot	575	10925
17	-	-	-	-	-	-	Unpaved parking Lot	7	139
18	-	-	-	-	-	-	Cars	327	6220
19	-	-	-	-	-	-	Trains	268	5101
20	-	-	-	-	-	-	Stadium seats	341	6483
	Total	1024	9225	Total	2138	40638	Total	25241	479615

encoder or decoder. In order to obtain a set of optimal learnable parameters, gradient descent algorithm is utilized during the search process to optimize the architecture parameters α, α' and α''. After the search, we retain the operators corresponding to the largest architecture parameter in each layer to construct a compact network.

Formally, we employ the bilevel optimization in [31] to jointly optimize the architecture parameters α, α', and α'' and network weights ω on the training and validation datasets

$$\begin{aligned} &\min_{\alpha} \mathcal{L}_{\text{val}}(\omega^*(\alpha), \alpha) + \lambda_e \mathcal{L}_{\text{Beta}} \\ &\text{s.t. } \omega^*(\alpha) = \arg\min_{\omega} \mathcal{L}_{\text{train}}(\omega, \alpha) \\ &\mathcal{L}_{\text{Beta}} = \log\left(\sum_{k=1}^{|O|} e^{\alpha_k}\right) = \text{smoothmax}(\{\alpha\}) \end{aligned} \tag{5}$$

where $\alpha \in \{\alpha, \alpha', \alpha''\}$ and according to [32], $\lambda_e = 50 \times (e/E), e \in \{1, 2, \ldots, E\}$ represents the weight of beta decay regularization loss (E represents the total number of epochs). After the bilevel optimization, the optimal learnable parameters are obtained. Then, we choose the specific operator for each layer of the U²ConvFormer according to the optimal parameters.

IV. EXPERIMENTS

In this section, we assess the effectiveness of the proposed U²ConvFormer for HSI classification. First, we present the datasets employed in our study. Then, we describe the experimental settings. Next, we compare our U²ConvFormer with other state-of-the-art CNN-based, ViT-based, and CNN-and-ViT-combined HSI classification methods. Furthermore, we conduct the ablation studies to analyze the effectiveness of each component in U²ConvFormer and the parameter settings.

A. Dataset Description

We use three challenging HSI benchmarks to evaluate the performance of the proposed U²ConvFormer, including Indian Pines, Pavia University, and Houston University 2018. We provide a concise overview of the three datasets. For comprehensive information, please refer to Table I.

1) Indian Pines: Indian pines is gathered by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor in 1992 over the Indian Pines test site in North-western Indiana, United States. It contains 145 × 145 pixels and 224 spectral reflectance bands in the wavelength range 0.4–2.5 μm. After removing the bands covering the region of water absorption (104–108, 150–163, and 220), 200 bands are finally used for the analysis. The dataset contains 10 249 labeled samples, which are classified into 16 categories of ground objects that occur in this sense containing two-thirds agriculture, and one-third forest or other natural perennial vegetation.

2) Pavia University: Pavia University is acquired by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor in 2003 over Pavia, northern Italy. It contains 610 × 340 pixels and 115 spectral reflectance bands with a geometric resolution of 1.3 m. After removing 12 noisy bands, 103 bands are finally used for the analysis. The dataset contains 42 776 labeled samples, which are classified into nine categories.

3) Houston University 2018: Houston University 2018 is gathered by the National Center for Airborne Laser Mapping (NCALM) at the University of Houston, over the University of Houston campus and its neighborhood on February 16, 2017. It contains 601 × 2384 pixels and 48 spectral reflectance bands in the wavelength range 380–1050 nm. The dataset contains 504 856 labeled samples, which are classified into 20 categories of land cover.

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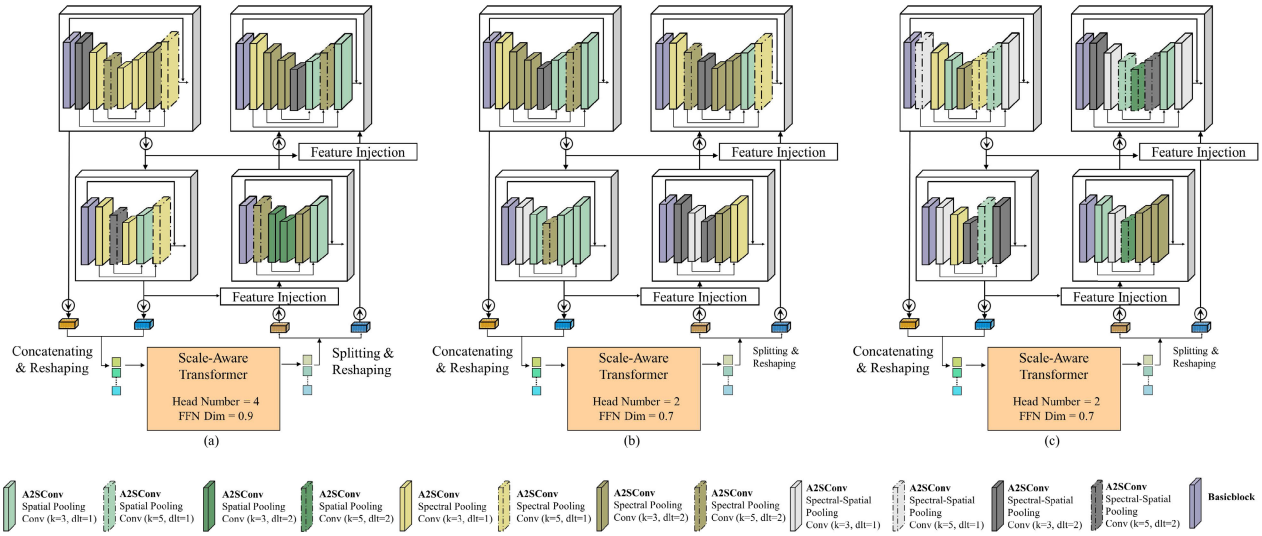


Fig. 5. Final network structures. (a) Indian Pines. (b) Pavia University. (c) Houston University 2018.

TABLE II
COMPARISON RESULTS OF DIFFERENT HSI CLASSIFICATION MODELS ON INDIAN PINES DATASET

Class No.	2-D CNN	3-D CNN	U-Net	SSRN	SF	SSFTT	GAHT	SSTN	3-D Auto CNN	HyT-NAS	U^{2}ConvFormer
OA(%)	87.77	85.42	96.81	97.75	92.31	97.47	97.95	98.03	95.26	98.30	**99.16**
AA(%)	86.06	75.10	73.68	94.71	84.95	96.57	97.75	97.54	87.63	96.39	**98.68**
Kappa	86.03	83.24	96.36	97.43	91.24	97.11	97.66	97.74	94.58	98.06	**99.04**
1	65.85	58.54	0.0	94.14	63.00	95.12	97.56	94.63	85.85	95.12	**100.00**
2	**99.77**	76.19	99.61	97.84	92.35	97.67	98.05	98.16	91.24	97.12	98.57
3	81.66	77.64	98.53	97.54	86.86	**98.87**	98.66	97.19	94.67	95.45	97.28
4	96.71	52.11	**100.00**	90.70	88.96	91.55	95.31	94.46	98.97	96.24	99.69
5	85.75	93.56	97.70	97.75	92.49	96.32	95.17	98.02	91.59	97.01	**99.08**
6	97.87	98.17	99.54	99.24	99.12	99.54	99.85	97.08	96.83	**100.00**	99.54
7	**100.00**	36.00	0.0	81.60	52.50	**100.00**	**100.00**	**100.00**	51.20	96.00	94.67
8	**100.00**	98.60	**100.00**	**100.00**	99.16	**100.00**	99.91	**100.00**	**100.00**	**100.00**	**100.00**
9	50.00	55.56	0.0	74.44	41.18	88.89	**100.00**	96.67	37.78	72.22	**100.00**
10	35.54	82.86	97.03	94.77	93.16	97.71	94.29	95.64	93.92	95.77	**99.58**
11	88.01	90.45	99.64	98.87	92.27	98.69	99.37	98.74	97.42	99.64	**99.82**
12	98.13	62.55	89.70	97.83	85.44	98.13	96.63	97.57	93.70	**98.69**	98.32
13	99.46	88.65	98.92	99.24	99.02	97.28	**100.00**	99.57	98.16	**100.00**	97.84
14	99.91	99.39	**100.00**	99.18	96.73	99.91	97.89	99.56	99.35	99.82	99.76
15	91.35	86.17	98.27	93.95	83.41	98.84	97.12	98.16	93.37	**99.14**	**99.14**
16	86.90	45.24	0.0	98.33	93.50	95.54	94.05	95.24	78.09	**100.00**	95.64

B. Comparison With State-of-the-Art Methods

In this section, we first introduce the experiment settings of the proposed U^{2}ConvFormer in the architecture searching stage and the network optimization stage. Then, we list the compared advanced methods and exhibit the comparison results.

1) Experimental Settings: The proposed U^{2}ConvFormer is searched, trained, and tested in the PyTorch framework. All experiment procedures, including the architecture search stage and the network optimization stage, are performed on one Tesla V100 GPU, with 32G memory.

a) Architecture searching: In order to find the optimal A2SConv operation, number of heads and MLP ratio, we train the supernet on each dataset for 100 epochs. The network architecture parameters (α, α', and α'') are optimized by the Adam optimizer and β-Decay Regularization, where learning rate and weight decay are set to 0.001 and 0.0003, respectively. Network parameters (ω) is also optimized by the Adam optimizer, where the momentum and weight decay are set to 0.9 and 0.0003, respectively. According to the step decay strategy, the learning rate decays from 0.001 to 0.0005 after the iteration of 500 epochs (i.e., half of the total epoch number). For the Indian Pines, Pavia University, and Houston University 2018, batch size is set to 8 and choose 30 labeled samples from each category to build training set and validation set. Fig. 5 shows the structures searched by the proposed method. Since the three datasets have different spectral and spatial resolutions as well as land cover, we searched for architectures on each dataset separately.

b) Network optimization: The same optimization strategy is applied to the three different networks searched on the three datasets. In terms of loss function, the cross-entropy loss is utilized. The Adam optimizer is used to optimize the searched network. Similar with the architecture searching phase, the momentum is set to 0.9 and weight decay is set to 0.0003. The training epoch is set to 50 for Pavia University, 100 for Indian Pines, and Houston University 2018, respectively. According to the step decay strategy, the learning rate decays to half of the current for every tenth of the total number, and the initial learning rate is set to 0.001. For the Indian Pines, Pavia University, and Houston University 2018, batch size is

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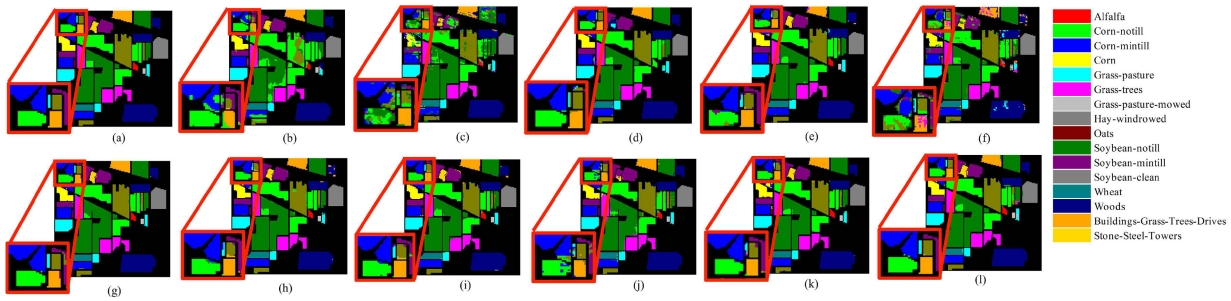


Fig. 6. Comparison results of HSI classification models on Indian Pines dataset. (a) Groundtruth. (b) Two-dimensional CNN, OA = 87.77%. (c) Three-dimensional CNN, OA = 85.42%. (d) U-Net, OA = 96.81%. (e) SSRN, OA = 97.75%. (f) SF, OA = 92.31%. (g) SSFTT, OA = 97.47%. (h) GAHT, OA = 97.95%. (i) SSTN, OA = 98.03%. (j) Three-dimensional Auto CNN, OA = 95.26%. (k) HyT-NAS, OA = 98.30%. (l) U²ConvFormer, OA = 99.16%.

TABLE III
COMPARISON RESULTS OF DIFFERENT HSI CLASSIFICATION MODELS ON PAVIA UNIVERSITY DATASET

Class No.	2-D CNN	3-D CNN	U-Net	SSRN	SF	SSFTT	GAHT	SSTN	3-D Auto CNN	HyT-NAS	U²ConvFormer
OA(%)	98.86	97.88	94.75	99.10	97.54	99.21	99.53	99.18	97.06	99.67	**99.76**
AA(%)	97.97	95.26	88.45	98.60	95.88	98.69	99.37	98.78	92.43	99.39	**99.53**
Kappa	98.48	97.19	96.62	98.81	96.74	99.15	99.37	98.92	96.10	99.56	**99.70**
1	99.68	97.02	99.90	98.81	96.21	99.33	99.38	98.78	97.79	99.46	**99.92**
2	99.41	99.97	99.89	99.83	99.64	99.92	99.80	99.97	99.88	**99.99**	99.97
3	86.56	92.98	**100.00**	92.45	87.65	98.29	98.35	97.30	96.80	96.09	99.92
4	98.18	97.53	96.67	98.32	96.64	98.49	99.52	97.21	82.90	**99.76**	98.47
5	99.84	99.06	**100.00**	99.65	99.97	99.53	**100.00**	**100.00**	94.83	**100.00**	99.92
6	**100.00**	99.10	**100.00**	99.43	99.56	**100.00**	99.75	99.92	99.99	**100.00**	**100.00**
7	99.84	79.10	**100.00**	99.76	90.3	99.13	99.60	97.73	99.14	**100.00**	99.52
8	**100.00**	97.34	99.66	99.32	94.6	98.05	98.63	97.24	98.53	99.74	99.53
9	98.22	95.22	0.00	**99.82**	98.49	95.44	99.33	98.82	62.00	99.44	98.56

set to 32, 32, and 64, respectively. The patch size utilized for the three datasets is set to 31. For the purpose of a fair comparison, we follow the settings of [58] in training and testing. We casually sample 10% of the total labeled samples from the Indian Pines, and 5% of the total labeled samples from the Pavia University and Houston University 2018 datasets as the training set. The specific figures can be found in Table I.

2) Competitive Approaches: We compare the proposed U²ConvFormer with nine outstanding methods. The compared methods can be classified into three types, which are the CNN-based methods 2-D CNN, 3-D CNN, U-Net and SSRN [59], ViT-based and CNN-and-ViT-combined networks spectralformer (SF) [6], SSFTT [8], GAHT [38] and SSTN [53], and NAS-based approaches 3-D Auto CNN [51], and HyT-NAS [55]. The details are listed as follows. Both 2-D CNN and 3-D CNN include three convolution layers and one softmax layer. For more specific information, please consult the original papers for detailed [8], [38], [51], [53], [55], [59]. For all experiments conducted on the compared approach, the parameters are set the same as the original paper. We use three widely used quality metrics—overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) to evaluate the appearance of different methods. We repeat each experiment with the same setup five times to ensure the equity and stability, and the average result is presented.

The quantitative results carried out on the Indian Pines dataset using the ten models can be found in Table II. The corresponding visualization results are presented in Fig. 6. The most exceptional results in each row are emphasized in bold. Apparently, the proposed U²ConvFormer achieves the highest classification values when compared to other comparable methods. In terms of Indian Pines dataset, the number of training samples of certain categories is limited, such as class 1, 7, and 9. U²ConvFormer also presents the best AA with the best classification accuracy on class 1 and 9.

The quantitative results carried out on the Pavia University dataset using the ten models can be found in Table III. The corresponding visualization results are presented in Fig. 7. The most exceptional results in each row are emphasized in bold. Apparently, the proposed U²ConvFormer achieves the highest classification values when compared to other comparable methods. Therefore, the classification performance of most methods behaves well. Among them, U²ConvFormer obtained favorable classification results on both narrow roads and buildings.

The quantitative results carried out on the Houston University 2018 dataset using the ten models can be found in Table IV. The corresponding visualization results are presented in Fig. 8. The most exceptional results in each row are emphasized in bold. Apparently, the proposed U²ConvFormer achieves the highest classification values when compared to other comparable methods. The dataset exhibits high resolution and intricate distributions of land cover, rendering it considerably more challenging to classify. By marrying and evolving the advantages of nested U-Net and scale-aware Transformer to extract spectral-spatial information, our U²ConvFormer obtains obvious superiority over other approach.

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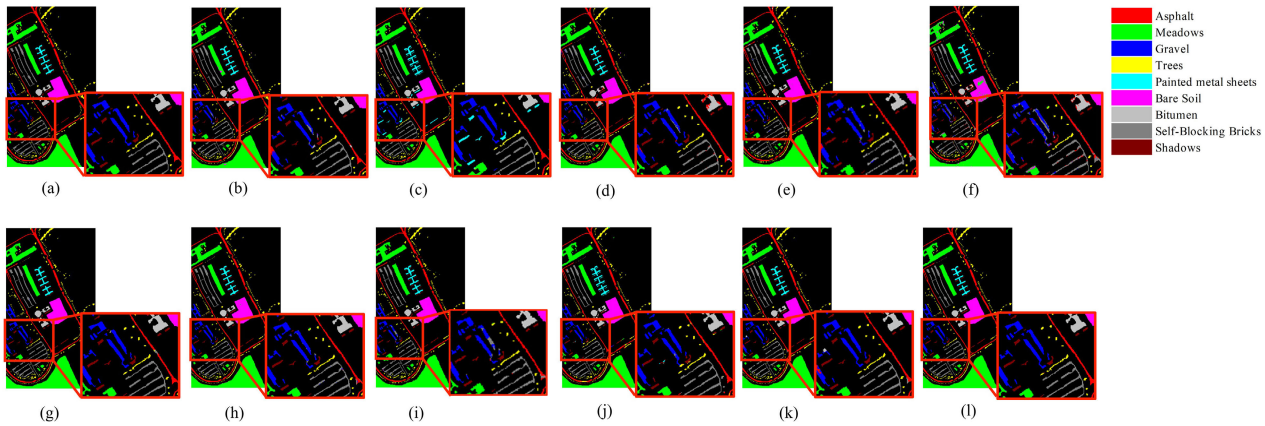


Fig. 7. Comparison results of HSI classification models on Pavia University dataset. (a) Groundtruth. (b) Two-dimensional CNN, OA = 98.86%. (c) Three-dimensional CNN, OA = 97.88%. (d) U-Net, OA = 96.81%. (e) SSRN, OA = 99.10%. (f) SF, OA = 97.54%. (g) SSFTT, OA = 99.21%. (h) GAHT, OA = 99.18%. (i) SSTN, OA = 99.53%. (j) Three-dimensional Auto CNN, OA = 97.06%. (k) HyT-NAS, OA = 99.67%. (l) U^{2}ConvFormer, OA = 99.76%.

TABLE IV
COMPARISON RESULTS OF DIFFERENT HSI CLASSIFICATION MODELS ON HOUSTON UNIVERSITY 2018 DATASET

Class No.	2-D CNN	3-D CNN	U-Net	SSRN	SF	SSFTT	GAHT	SSTN	3-D Auto CNN	HyT-NAS	U^{2}ConvFormer
OA(%)	94.84	86.88	94.60	91.61	90.50	95.48	96.69	90.25	85.45	97.12	**97.70**
AA(%)	88.79	80.50	75.71	82.63	79.86	90.26	92.85	83.79	70.13	94.93	**95.38**
Kappa	93.25	83.13	92.96	89.06	87.65	94.12	95.70	87.29	80.90	96.25	**97.01**
1	84.33	82.02	84.82	86.30	**94.69**	79.93	79.50	75.23	61.88	82.29	87.26
2	95.98	96.64	96.45	95.32	93.90	93.44	96.55	95.71	86.39	**98.49**	96.08
3	96.92	96.00	**100.00**	99.72	0.00	99.66	**100.00**	97.57	91.51	99.85	99.33
4	98.13	95.55	97.56	97.49	98.78	96.64	97.62	95.82	87.09	98.61	**98.87**
5	91.26	79.16	92.70	86.09	90.16	90.11	95.01	74.77	47.62	**97.59**	95.74
6	99.91	97.51	**100.00**	98.48	99.43	99.62	99.91	98.74	97.90	**100.00**	99.97
7	80.63	71.94	77.87	93.68	40.52	85.45	95.65	98.66	47.52	**100.00**	99.87
8	94.29	88.62	**99.78**	91.62	96.05	98.72	99.18	91.84	97.32	99.24	99.75
9	99.54	92.80	**99.57**	97.88	98.96	99.09	99.35	97.75	98.11	99.28	99.53
10	**95.44**	71.61	94.21	81.72	88.34	91.15	92.64	73.62	58.07	90.59	94.92
11	77.86	73.17	86.71	69.44	23.79	80.97	85.73	62.69	31.19	89.23	**90.16**
12	15.26	11.10	48.31	0.51	31.90	41.69	33.43	5.57	0.28	**53.10**	52.94
13	87.30	69.12	97.70	84.59	91.30	94.49	96.79	86.76	90.87	**97.89**	97.85
14	94.97	96.18	99.32	89.65	94.84	96.97	99.17	94.32	92.00	98.28	**99.45**
15	95.40	**98.98**	99.86	99.34	99.39	99.30	99.92	98.22	86.67	99.83	**99.98**
16	93.62	90.40	**99.56**	91.00	96.03	97.84	98.15	91.61	86.01	99.24	99.45
17	86.33	20.86	58.99	0.00	0.00	69.21	90.65	54.82	0.00	**97.84**	97.60
18	96.64	89.05	**99.39**	93.66	91.06	93.07	97.85	87.95	66.67	97.64	98.92
19	99.24	95.45	99.75	96.92	97.44	97.97	99.84	79.63	96.67	99.63	**99.93**
20	92.77	93.92	**100.00**	99.12	99.84	99.96	**100.00**	97.53	96.77	**100.00**	99.97

C. Ablation Study

This section analyzes in detail the impact of each component proposed in our U^{2}ConvFormer. The results are reported on Indian Pines, Pavia University, and Houston University 2018. The settings for the searched model during training and testing are the same as in Section IV-B.

1) Ablation for the Encoder and Decoder Number: Our U^{2}ConvFormer architecture consists of two parts: a well-designed U^2-Net that includes an m-layer encoder and an m-layer decoder, and a scale-aware Transformer. As can be expected, different values of m will affect the performance of the final searched model. The classification result (measured by OA) of m taking a set of {1, 2, 3} is shown in Fig. 9. From the figure, we can see that the classification is best on Indian Pines when m is set to 2. On the other two datasets, the advantage of U^{2}ConvFormer with a two-layer encoder and a two-layer decoder in classification accuracy gradually appears with the increase of patch size. Combining the performance on the three datasets and the effect of patch size on the classification results (please refer to Section IV-D), we choose to set m to 2 in this article. On the one hand, when m is taken as 1, the structure of the model is too shallow to extract the deep spectral-spatial features. On the other hand, when m is taken as 3, the model has too many parameters leading to overfitting. At the same time, it also causes a decrease in computational speed.

2) Ablation for A2SConv Module: As mentioned in Section III-B, asymmetric spectral-spatial feature pooling and multiscale feature extraction can be concurrently searched in

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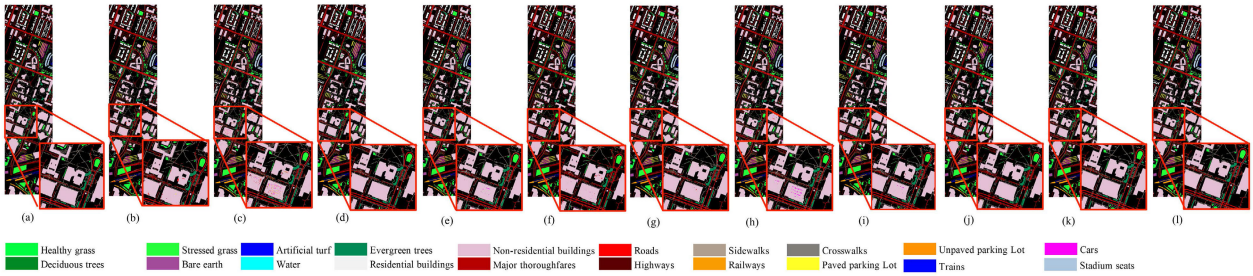


Fig. 8. Comparison results of HSI classification models on Houston University 2018 dataset. (a) Groundtruth. (b) Two-dimensional CNN, OA = 94.84%. (c) Three-dimensional CNN, OA = 86.88%. (d) U-Net, OA = 97.21%. (e) SSRN, OA = 91.61%. (f) SF, OA = 90.50%. (g) SSFTT, OA = 95.48%. (h) GAHT, OA = 96.69%. (i) SSTN, OA = 90.25%. (j) Three-dimensional Auto CNN, OA = 85.45%. (k) HyT-NAS, OA = 97.12%. (l) U²ConvFormer, OA = 97.70%.

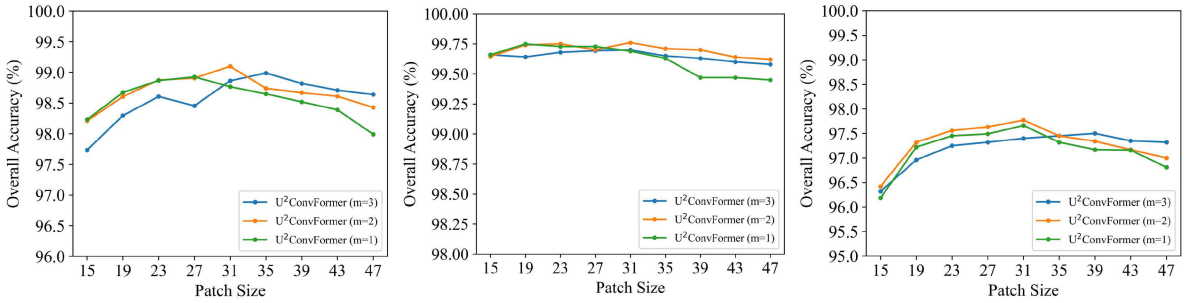


Fig. 9. Classification accuracy in terms of OA by varying the patch size on three HSI datasets. (a) Indian Pines. (b) Pavia University. (c) Houston University 2018.

A2SConv. To verify the effect of A2SConv in U²ConvFormer, we replaced it with a 3-D convolution block with kernel size of $3 \times 3 \times 3$. The rest of the experimental design remains the same. The new results are shown in Table V. It is seen that the U²ConvFormer composed of A2SConv is always better than U²ConvFormer composed of plain 3-D CNN with decrease of model parameters since it can maximally retains the essential spectral-spatial features of HSI.

3) Ablation for Scale-Aware Transformer Module: The scale-aware Transformer, mentioned in Section III-B, takes the multiscale local spectral-spatial features as input and produces scale-aware global spectral-spatial features. To verify the effect of scale-aware Transformer module in U²ConvFormer, we remove it and conduct experiments with the same setups as before. The new results are shown in Table V. As can be seen, the utilization of scale-aware Transformer module brings gains in OA, AA, and Kappa for Indian Pines, Pavia University, and Houston University 2018.

4) Ablation for Feature Injection Module: The feature injection module, mentioned in Section III-B, injects the multiscale global spectral-spatial features with the scale-aware local spectral-spatial features to augment the representation of HSI cubes. To verify the effect of the feature injection module in U²ConvFormer, we remove it and conduct experiments with the same setups as before. The results are shown in Table V. As can be seen, the utilization of feature injection module brings gains in OA, AA, and Kappa for Indian Pines, Pavia University, and Houston University 2018 datasets.

D. Parameter Analyses

This section analyzes the impact of patch size and the amount of training samples on the proposed U²ConvFormer in detail. The results are reported on the three datasets. The settings for the searched model during training and testing are the same as in Section IV-B.

1) Effect of Different Patch Sizes: In order to investigate the impact of patch size, the ablation studies of patch size are conducted, and the results (measured by OA) are shown in Fig. 9. The input patch size is set from 15×15 to 47×47, where the stride size is set to 4. On one hand, the classification accuracy increases with the increase of patch size when the patch size is relatively small, while the classification accuracy decreases with the increase of patch size when the patch size is large. The reason may be that small patch size can capture the local details while lack of utilization of global information, and large ones can better represent the global features while failing to distinguish the subtle local differences. On the other hand, the classification accuracy is affected by the searched results of the nested U-Net structure of U²ConvFormer. Further, in individual cases on the three datasets, the classification results fluctuates as the increase of patch size. The reason why the irregular results occurred in Pavia University dataset may be that the results of our method on this dataset are always over 99%, so there is no significant difference for different patch sizes.

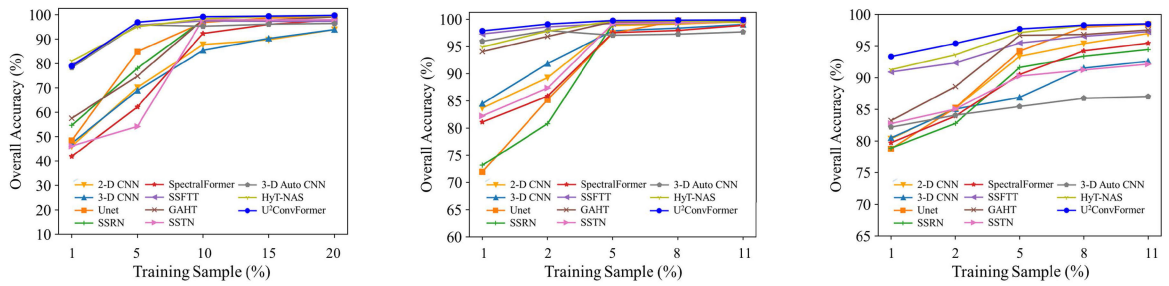
2) Effect of Different Number of Training Samples: In Fig. 10, we show the results under various proportions of training sets. All comparable methods are trained utilizing 1%, 5%, 10%, 15%, and 20% of the samples as the training set for Indian Pines dataset, as well as 1%, 2%, 5%, 8%, and 11% of the samples as the training set for Pavia University dataset and Houston University 2018 dataset, respectively. As shown in Fig. 10, the proposed U²ConvFormer method achieves the best OA on Pavia University dataset and Houston University 2018 dataset with 1%, 2%, 5%, 8% and 11% of the samples as the training set, as well as on the Indian Pine dataset

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TABLE V

ABLATION STUDY OF A2SCONV, SCALE-AWARE TRANSFORMER AND FEATURE INJECTION ON THE THREE HSI DATASETS. THE MOST EXCEPTIONAL RESULTS IN EACH ROW ARE EMPHASIZED IN BOLD

Dataset	Method		Params (M)	Flops (M)	OA (%)	AA (%)	Kappa
Indian Pines	U^2ConvFormer w/o	A2SConv and searchable Transformer	0.22	4518.78	95.17	84.34	94.48
		searchable Transformer	0.61	843.65	97.77	86.74	97.45
		A2SConv replaced by Conv3d (k=3)	0.32	4503.30	98.22	93.20	97.97
		A2SConv replaced by Conv3d (k=5)	1.34	12724.11	98.30	93.11	98.06
		feature injection	0.26	501.83	98.50	96.74	98.29
	U^2ConvFormer		0.71	859.13	**99.16**	**98.68**	**99.04**
Pavia University	U^2ConvFormer w/o	A2SConv and searchable Transformer	0.22	1355.15	99.16	98.39	98.89
		searchable Transformer	0.32	329.75	99.30	98.68	99.07
		A2SConv replaced by Conv3d (k=3)	0.58	1326.85	99.66	99.30	99.55
		A2SConv replaced by Conv3d (k=5)	2.19	9383.67	99.67	99.17	99.57
		feature injection	0.39	254.52	98.50	96.74	98.29
	U^2ConvFormer		0.41	358.05	**99.76**	**99.53**	**99.70**
Houston University 2018	U^2ConvFormer w/o	A2SConv and searchable Transformer	0.22	3149.69	97.12	93.91	96.36
		searchable Transformer	0.33	291.59	97.48	94.17	96.71
		A2SConv replaced by Conv3d (k=3)	0.58	3451.42	98.22	93.20	97.97
		A2SConv replaced by Conv3d (k=5)	2.19	8756.29	97.45	94.28	96.69
		feature injection	0.41	231.76	97.40	95.00	96.63
	U^2ConvFormer		0.41	323.32	**97.70**	**95.38**	**97.01**



(a) (b) (c)

Fig. 10. Classification accuracy in terms of OA by varying the numbers of the training samples on three HSI datasets. (a) Indian Pines. (b) Pavia University. (c) Houston University 2018.

with 5%, 10%, 15% and 20% samples as training set. The proposed U^2ConvFormer method achieves suboptimal OA on the Indian Pines dataset with 1% samples as training set. This is because several categories in the Indian Pines dataset have less than 100 samples. When 1% of the samples are used as the training set, only one sample can be used for training, and the performance of the proposed method is limited in this case.

V. CONCLUSION

In this article, we propose the novel architecture, namely U^2ConvFormer, for HSI classification. The nested U-Net structure can fully extract and aggregate multiscale spectral-spatial features at both inter- and inner-stage granularity. The scale-aware Transformer takes multiscale local spectral-spatial features from the encoder of nested U-Net and produces

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multiscale global spectral-spatial features for its decoder. To the best of our knowledge, it is the first time to marry and evolve the advantages of nested U-Net and scare-aware Transformer to construct a model to classify HSI data. Moreover, we design a novel lightweight and plug-and-play searchable operation called A2SConv, which can be flexibly built into U^{2}ConvFormer, where asymmetric spectral-spatial feature pooling and multiscale feature extraction can be concurrently searched. Furthermore, we develop a customized search strategy to automatically design U^{2}ConvFormer, which uses advanced NAS methods to enable the customization of suitable models for different hyperspectral datasets. Extensive experimental results on three HSI datasets validate the superiority of our proposed U^{2}ConvFormer, which achieves new state-of-the-art performance across different benchmark datasets.

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